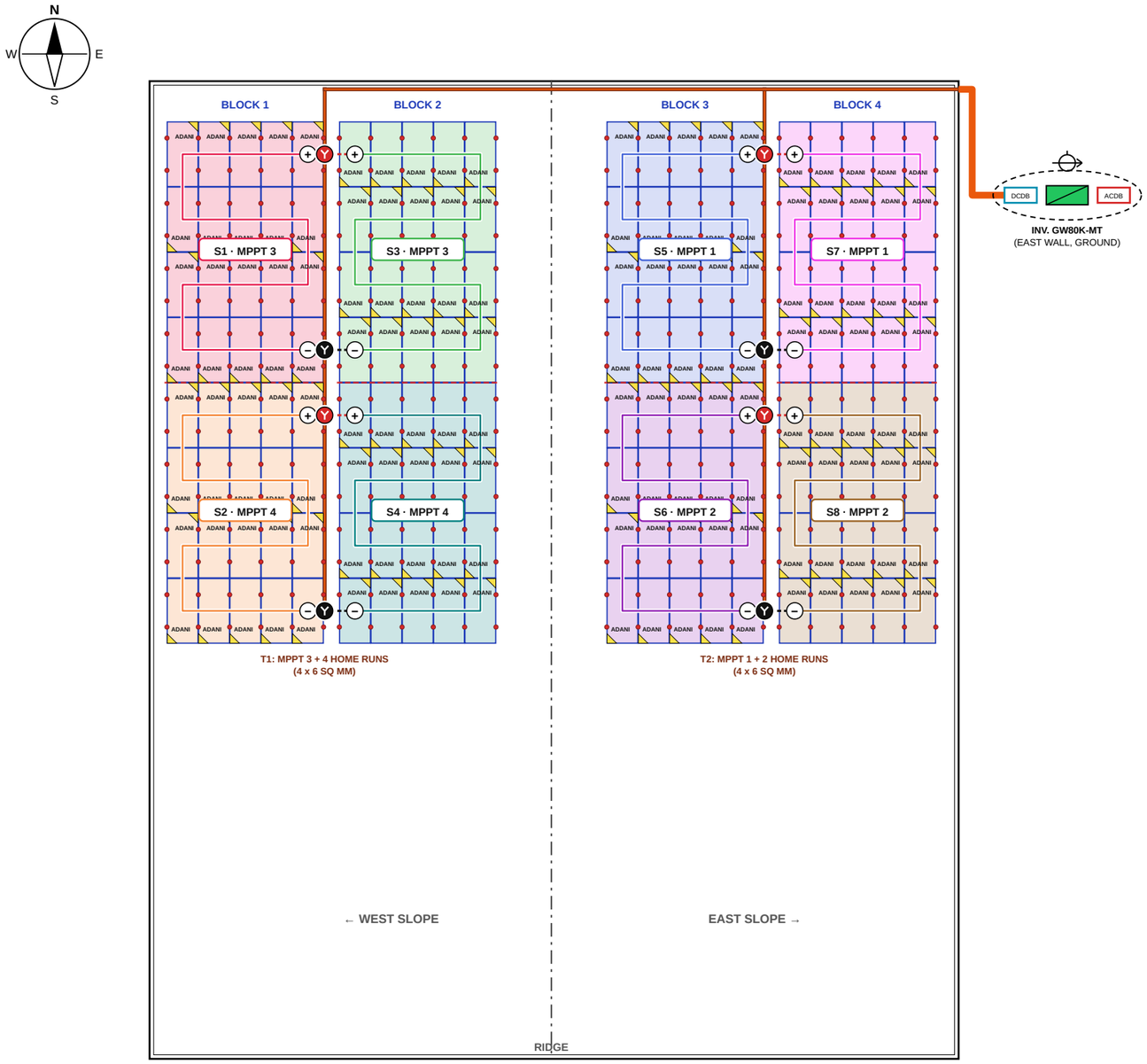
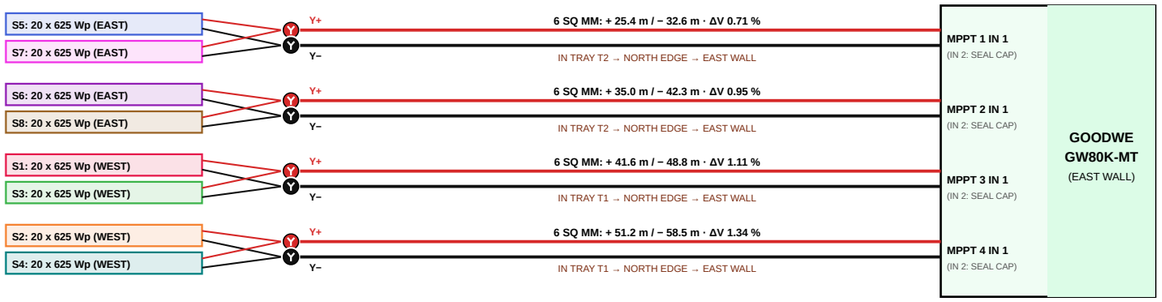


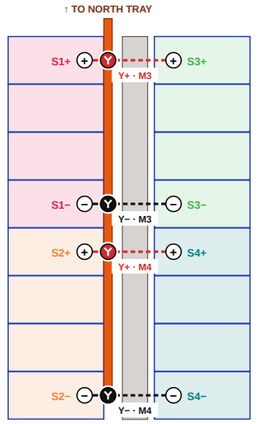
ELECTRICAL WIRING DIAGRAM — STRINGS → Y-CONNECTORS → 6 SQ MM HOME RUN → GW80K-MT



NOTE: OWNER'S MIRRORED SNAKE — + BESIDE + AND − BESIDE − ACROSS EACH GAP, SO THE TWO PARTNER STRINGS OF AN MPPT ARE JOINED RIGHT THERE BY ONE + Y AND ONE − Y. ONE 6 SQ MM +/− PAIR PER MPPT RUNS IN THE TRAY TO THE INVERTER. EAST AND WEST NEVER SHARE AN MPPT / Y. UNUSED INVERTER INPUTS (IN 2 OF EACH MPPT) GET THE SUPPLIED SEALING CAPS. INVERTER NOW MONITORS PER MPPT, NOT PER STRING. LENGTHS SCALED FROM THE LAYOUT IMAGE (*) — VERIFY ON SITE.

DETAIL A — Y-CONNECTION IN GAP 1/2

(GAP 3/4 IDENTICAL: S5/S7 = MPPT 1, S6/S8 = MPPT 2)



NEAR STRING (S1 / S2): MODULE LEAD PLUGS STRAIGHT INTO THE Y. FAR STRING (S3 / S4): 1.5 m 4 SQ MM LEAD (DASHED), CLIPPED UNDER THE W1 GRATING, TO THE Y. ROWS STRETCHED ACROSS THE SLOPE.

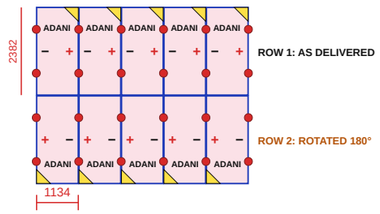
STRING DESIGN CHECK

MODULES / STRING	20 (ADANI 625 Wp N-TOPCon)
Voc @ 5 °C	20 x 52.3 = 1,046 V < 1,100 V ✓
Vmp RANGE	723 – 880 V (MPPT 200 – 1,000 V) ✓
CURRENT / MPPT	Y = 2 STRINGS: Imp 30.1 A ≤ 44 A ; Isc 31.9 ≤ 55 A ✓
Y-CONNECTOR	≥ 40 A, 1500 V DC (1.25 x 31.9 A = 39.9 A)
HOME RUN	6 SQ MM, ONE +/− PAIR PER MPPT
V-DROP (6 SQ MM)	0.71 – 1.34 % (≤ 1.5 %) ✓
STRING FUSE	NOT REQUIRED (2 STRINGS / MPPT)

⚠ MANY STANDARD MC4 Y-BRANCHES ARE RATED ONLY 30 A — BELOW THE 30.1 A THEY WILL CARRY HERE. BUY ≥ 40 A, SAME MAKE AS THE MODULE CONNECTORS (MIXED MAKES = HOT JOINTS).

DETAIL B — PANEL PLACING (S1, ROWS 1–2, BLOCK 1)

LANDSCAPE: LONG SIDE N–S, ALONG THE RIDGE



● = MOUNTING SET (80 x 100 TRAPEZOIDAL SECTION) ON EVERY LONG-EDGE LINE, AT ¼ AND ¾ OF THE MODULE LENGTH (≈ 595 MM FROM EACH END). INNER SETS CLAMP TWO MODULES (MID CLAMP), END SETS ONE (END CLAMP).
◥ = ADANI LABEL CORNER, ON THE MODULE'S + SIDE: TOP RIGHT (NE) AS DELIVERED IN ROWS 1, 3, 5, 7 OF BLOCKS 1 & 3; BOTTOM LEFT (SW) = MODULE ROTATED 180° IN THE OTHER ROWS (SEE PLAN FOR EACH BLOCK). MOUNTING SETS ON THIS ROOF: 384 NOS (4 BLOCKS x 8 ROWS x 6 LINES x 2).

LEGENDS

	MODULE (TINT = STRING)
	ADANI LABEL CORNER (NE / SW = ROTATED 180°)
	MOUNTING SET (80 x 100 SECTION)
	STRING WIRING ORDER
	STRING + / − TERMINAL
	Y-CONNECTOR + / − (≥ 40 A)
	FAR-STRING LEAD 4 SQ MM
	DC CABLE TRAY 50 x 50 MM
	ROOF RIDGE
	STRING INVERTER

SYSTEM DESCRIPTION

SITE LOCATION	— (LAT, LONG TO BE ADDED)
DC CAPACITY	100.00 kWp
TOTAL PLANT CAPACITY (AC)	80 kW
DC / AC RATIO	1.25
MODULE CAPACITY	625 Wp (ADANI N-TOPCon)
MODULE DIMENSION (L X W X H)	2382 X 1134 X 30 mm *
TOTAL PV MODULES	160 Nos.
INVERTER CAPACITY	80 kW (GOODWE GW80K-MT) 1 No.
AZIMUTH / TILT	EAST-WEST (90° / 270°), ROOF SLOPE

CUSTOMER SIGN AND STAMP:

REV.	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	APPROVED BY
R0	29-10-26	FIRST ISSUE			

CLIENT:	**JAIN GRANITE & MARBLE**
EPC CONTRACTOR:	**NRG TECHNOLOGISTS PVT LTD** 989/6, GIDC Makarpura, Vadodara
DESIGN BY:	**NRG TECHNOLOGISTS PVT LTD** 989/6, GIDC Makarpura, Vadodara

DRAWING NO.: L-01

DRAWING NAME: **PANEL LAYOUT + ELECTRICAL WIRING (Y)**

PROJECT TITLE: 100.00 kWp (DC)_ROOFTOP SOLAR PLANT

RELEASED FOR: ☐ PRELIMINARY ☐ TENDER ☐ REVISION ☐ FABRICATION ☐ CONSTRUCTION ■ APPROVAL

SCALE: **1:200 (A3)**

DESIGN BY	DRAWN BY	CHECKED BY	APPROVED BY	PAPER SIZE	SHEET NO.
29-10-26	29-10-26	29-10-26	29-10-26	**A3**	**01 OF 03**